Please check the examination details below before entering your candidate information

Candidate surname	Other names

Group

Candidate Number

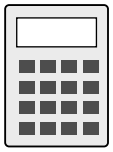
Elite IGCSE Academy | International GCSE

Expected Paper

November 2026

Time: **2 hours**

Paper reference **ELT/U1X**



Calculator allowed

Mathematics A

MODULAR · UNIT 1

Higher Tier · Expected Exam

You must have: Ruler graduated in centimetres and millimetres, protractor, pair of compasses, pen, HB pencil, eraser, calculator.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, group and candidate number.
- Answer **all** questions.
- Without sufficient working, correct answers may be awarded no marks.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- **Calculators may be used.**
- You must **NOT** write anything on the formulae page.

Information

- The total mark for this paper is 100.
- There are **26** questions in this paper.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Questions 19 to 26 are the most demanding. Aim to reach them with at least 45 minutes left.
- Check your answers if you have time at the end.

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Dr Eslam Ahmed
Tel. **01120009622**
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ELITE IGCSE ACADEMY

ELT U1X 2026

International GCSE Mathematics

Formulae sheet – Higher Tier

Arithmetic series

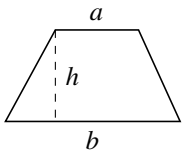
Sum to n terms, $S_n = \frac{n}{2}[2a + (n-1)d]$

The quadratic equation

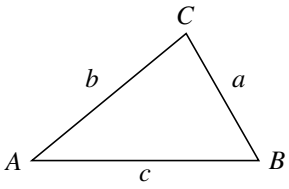
The solutions of $ax^2 + bx + c = 0$ where $a \neq 0$ are given by:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Area of trapezium $= \frac{1}{2}(a+b)h$



Trigonometry



In any triangle ABC

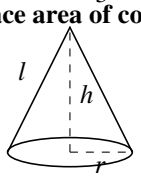
Sine Rule $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

Cosine Rule $a^2 = b^2 + c^2 - 2bc\cos A$

Area of triangle $= \frac{1}{2}ab\sin C$

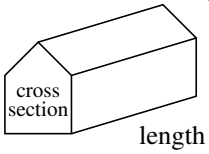
Volume of cone $= \frac{1}{3}\pi r^2 h$

Curved surface area of cone $= \pi r l$



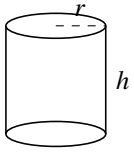
Volume of prism

= area of cross section × length



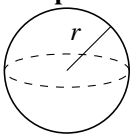
Volume of cylinder $= \pi r^2 h$

Curved surface area of cylinder $= 2\pi r h$



Volume of sphere $= \frac{4}{3}\pi r^3$

Surface area of sphere $= 4\pi r^2$



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Answer ALL TWENTY SIX questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

1 Without using a calculator, show that $\left(2\frac{1}{3} + 1\frac{4}{5}\right) \div 3\frac{1}{10} = 1\frac{1}{3}$

Show each stage of your working.

(Total for Question 1 is 3 marks)

2 By rounding each number to one significant figure, work out an estimate for the value of

$$\frac{\sqrt{48.7 \times 2.13}}{0.512}$$

Show your working clearly.

.....

(2)

(Total for Question 2 is 2 marks)

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3 $\mathscr{E} = \{1, 2, 3, \ldots, 24\}$

$A = \{\text{factors of } 24\}$ $B = \{\text{odd numbers}\}$ $C = \{\text{square numbers}\}$

(a) List the members of the set $A \cap B'$

.....

(1)

D is a set with exactly 3 members such that $D \subset A$, $D \cap B = \varnothing$ and $D \cap C = \{4\}$

(b) List the members of one possible set D.

.....

(2)

(Total for Question 3 is 3 marks)

---

4 Simplify fully $\dfrac{(2x^3y^{-1})^4}{4xy^2}$

.....

(2)

(Total for Question 4 is 2 marks)

---

5 In a bag there are 12 green counters, 5 blue counters and 3 red counters only.
Ben puts some extra blue counters and some extra red counters into the bag.
The number of blue counters he adds is **three times** the number of red counters he adds.

A counter is now taken at random from the bag. The probability that the counter is green is $\frac{3}{8}$

How many blue counters did Ben add to the bag?

.....

(3)

(Total for Question 5 is 3 marks)

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6 Solve $\left(\frac{1}{9}\right)^{x} \times 27^{x+2} = \sqrt{3}$

Show clear algebraic working.

$x =$ .....

(3)

(Total for Question 6 is 3 marks)

---

7 A solid cuboid measures 2 m by 1.5 m by 0.8 m.

When the cuboid rests on a face that measures 2 m by 1.5 m, the pressure on the floor is 400 newtons/m^2.

The cuboid is turned so that it rests on its **smallest** face.

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the pressure on the floor now.

..... newtons/m^2

(3)

(Total for Question 7 is 3 marks)

---

8 Expand and simplify $(2x - 3)^2(x + 2)$

.....

(3)

(Total for Question 8 is 3 marks)

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9 ABC and ACD are right-angled triangles.

$AB = 8.5$ cm

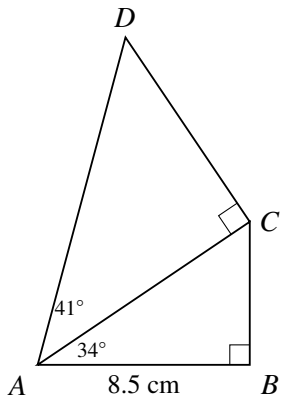
angle $ABC = 90°$ angle $BAC = 34°$

angle $ACD = 90°$ angle $CAD = 41°$

Work out the length of CD.

Give your answer correct to 3 significant figures.

Diagram **NOT** accurately drawn



. . . . . cm

(4)

(Total for Question 9 is 4 marks)

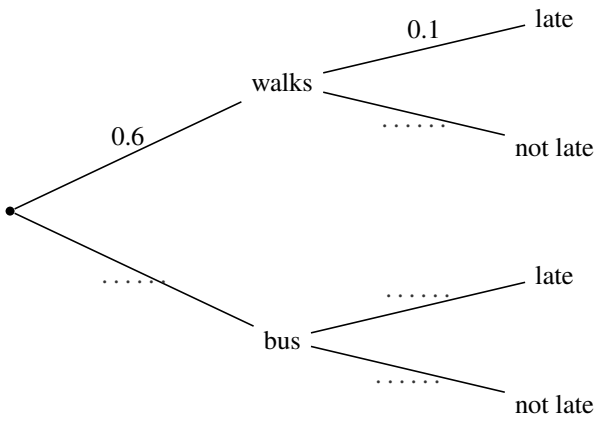
10 Each day Leila either walks to school or takes the bus.

The probability that she walks is 0.6

If she walks, the probability that she is late is 0.1

If she takes the bus, the probability that she is late is 0.25

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that Leila is late on a day chosen at random.

. . . . . .

(2)

(Total for Question 10 is 4 marks)

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11 Use algebra to show that the recurring decimal $1.2\dot{3}\dot{6} = \frac{68}{55}$

(Total for Question 11 is 2 marks)

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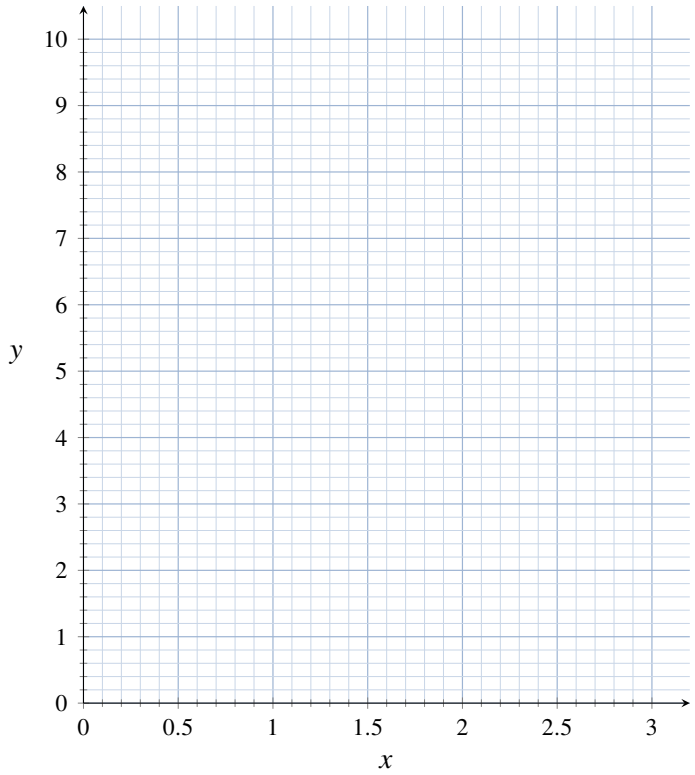
12

(a) Complete the table of values for $y = x^2 + \frac{2}{x}$

x	0.5	1	1.5	2	2.5	3
y	4.25		3.58		7.05	

(2)

(b) On the grid, draw the graph of $y = x^2 + \frac{2}{x}$ for $0.5 \leqslant x \leqslant 3$



(2)

(c) By drawing a suitable straight line on the grid, find estimates for the solutions of $x^3 - 3x^2 + 2 = 0$ in the range $0.5 \leqslant x \leqslant 3$

. . . . . .

(3)

(Total for Question 12 is 7 marks)

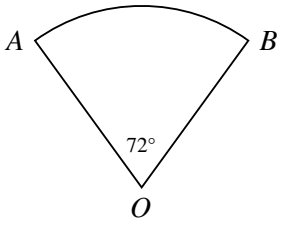
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13

OAB is a sector of a circle with centre O.
Angle $AOB = 72°$
The area of the sector is 20π cm^2

Find the perimeter of the sector.
Give your answer in the form $a + b\pi$ where a and b are integers.

Diagram **NOT** accurately drawn



. . . . . cm

(4)

(Total for Question 13 is 4 marks)

14

$$k = \frac{p - q}{r}$$

$p = 64$ correct to the nearest integer
$q = 23.6$ correct to 1 decimal place
$r = 1.4$ correct to 1 decimal place

By considering bounds, work out the value of k to a suitable degree of accuracy.
You must show all your working and give a reason for your answer.

$k =$ . . . . . .

(4)

(Total for Question 14 is 4 marks)

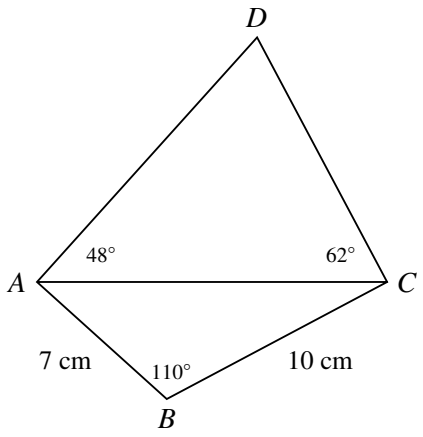
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15 The diagram shows quadrilateral $ABCD$.

$AB = 7\,\text{cm}$ $BC = 10\,\text{cm}$
angle $ABC = 110°$
angle $CAD = 48°$ angle $ACD = 62°$

Work out the length of AD.
Give your answer correct to 3 significant figures.

Diagram **NOT** accurately drawn



. . . . . cm

(5)

(Total for Question 15 is 5 marks)

---

16 A rectangle has area $\left(17 + 7\sqrt{5}\right)\text{cm}^2$ and length $\left(3 + \sqrt{5}\right)\text{cm}$.

Work out the width of the rectangle.
Give your answer in the form $\left(a + b\sqrt{5}\right)$ cm where a and b are integers.
Show each stage of your working clearly.

. . . . . cm

(3)

(Total for Question 16 is 3 marks)

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17 Express $\dfrac{x^2 - 9}{2x^2 + 5x - 3} \div \dfrac{x^2 - 3x}{4x^2 - 1}$ as a single fraction in its simplest form.
Show clear algebraic working.

. . . . . .

(4)

(Total for Question 17 is 4 marks)

18

(a) Factorise $6x^2 - 7x - 20$

. . . . . .

(2)

(b) Hence, solve $6x^2 - 7x - 20 = 0$

. . . . . .

(1)

(Total for Question 18 is 3 marks)

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19

(a) Express $2x^2 + 8x + 11$ in the form $a(x + b)^2 + c$ where a, b and c are integers.

.....

(2)

(b) Hence, write down the maximum value of $\dfrac{10}{2x^2 + 8x + 11}$ and the value of x at which it occurs.

maximum value .....

x = ...

(2)

(Total for Question 19 is 4 marks)

---

20 Solve $x\sqrt{3} + 4 = 2x + \sqrt{3}$

Give your answer in the form $a + b\sqrt{3}$ where a and b are integers.
Show each stage of your working clearly.

x = ...

(4)

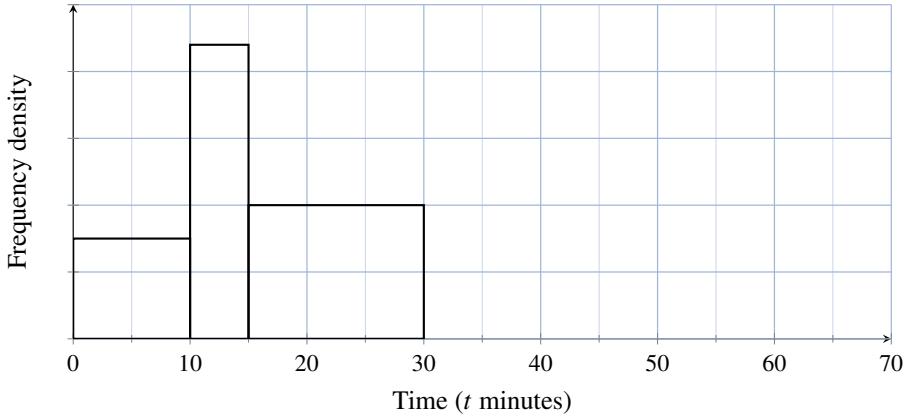
(Total for Question 20 is 4 marks)

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21 The incomplete histogram and the information below are about the times, in minutes, that 85 people spent at a gym. **No scale is shown on the vertical axis.**

- 15 people spent between 0 and 10 minutes.
- The bar for the class $30 < t \leqslant k$ is missing. Nobody spent more than k minutes.
- The frequency density of the missing class is 0.6



(a) Find the value of k and complete the histogram.

$k =$ .....

(3)

(b) Work out an estimate for the fraction of the 85 people who spent more than 25 minutes at the gym.

.....

(2)

(Total for Question 21 is 5 marks)

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22 $ABCD$ is a rhombus.
A is the point $(1, 2)$ and C is the point $(7, 10)$
The vertex B lies on the y-axis.

(a) Find an equation of the line BD.
Give your answer in the form $ax + by = c$ where a, b and c are integers.

.....

(3)

(b) Find the coordinates of D.

(..... , ...)

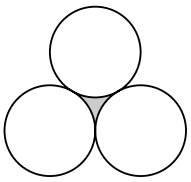
(2)

(Total for Question 22 is 5 marks)

---

23 Three identical circles, each of radius r cm, touch each other as shown in the diagram.

Diagram **NOT** accurately drawn



The region between the three circles is shaded.
The area of the shaded region is $10\,\text{cm}^2$

Work out the value of r.
Give your answer correct to 3 significant figures.

$r =$ .....

(4)

(Total for Question 23 is 4 marks)

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24 A bag contains 2 red beads, 3 blue beads and n green beads only.
Asha takes at random two beads from the bag without replacement.

The probability that the two beads are the **same colour** is $\frac{7}{15}$

(a) Show that $4n^2 - 39n - 10 = 0$

(3)

(b) Work out the number of green beads in the bag.

. . . . . .

(2)

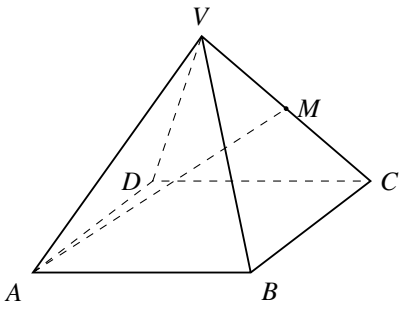
(Total for Question 24 is 5 marks)

---

25 $VABCD$ is a right pyramid.
The base $ABCD$ is a horizontal square of side 8 cm.
The vertex V is vertically above the centre of the base.
Each sloping edge of the pyramid has length 9 cm.
M is the midpoint of VC.

Work out the size of the angle between the line AM and the base $ABCD$.
Give your answer correct to one decimal place.

Diagram **NOT** accurately drawn



. . . . . °

(5)

(Total for Question 25 is 5 marks)

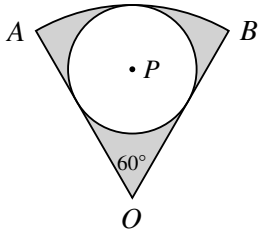
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26 The diagram shows a sector OAB of a circle, centre O.

Angle $AOB = 60°$

A circle, centre P and radius r cm, touches OA, OB and the arc AB.

Diagram **NOT** accurately drawn



(a) Show that the radius of the sector is $3r$ cm.

(2)

The area of the shaded region is 6π cm^2

(b) Work out the perimeter of the sector OAB.
Give your answer in the form $a\sqrt{3} + b\pi\sqrt{3}$ where a and b are integers.
Show your working clearly.

. . . . . cm

(4)

(Total for Question 26 is 6 marks)

TOTAL FOR PAPER IS 100 MARKS

For marking use

Question	1	2	3	4	5	6	7	8	9	10	11	12	13
Mark													
Question	14	15	16	17	18	19	20	21	22	23	24	25	26
Mark													

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